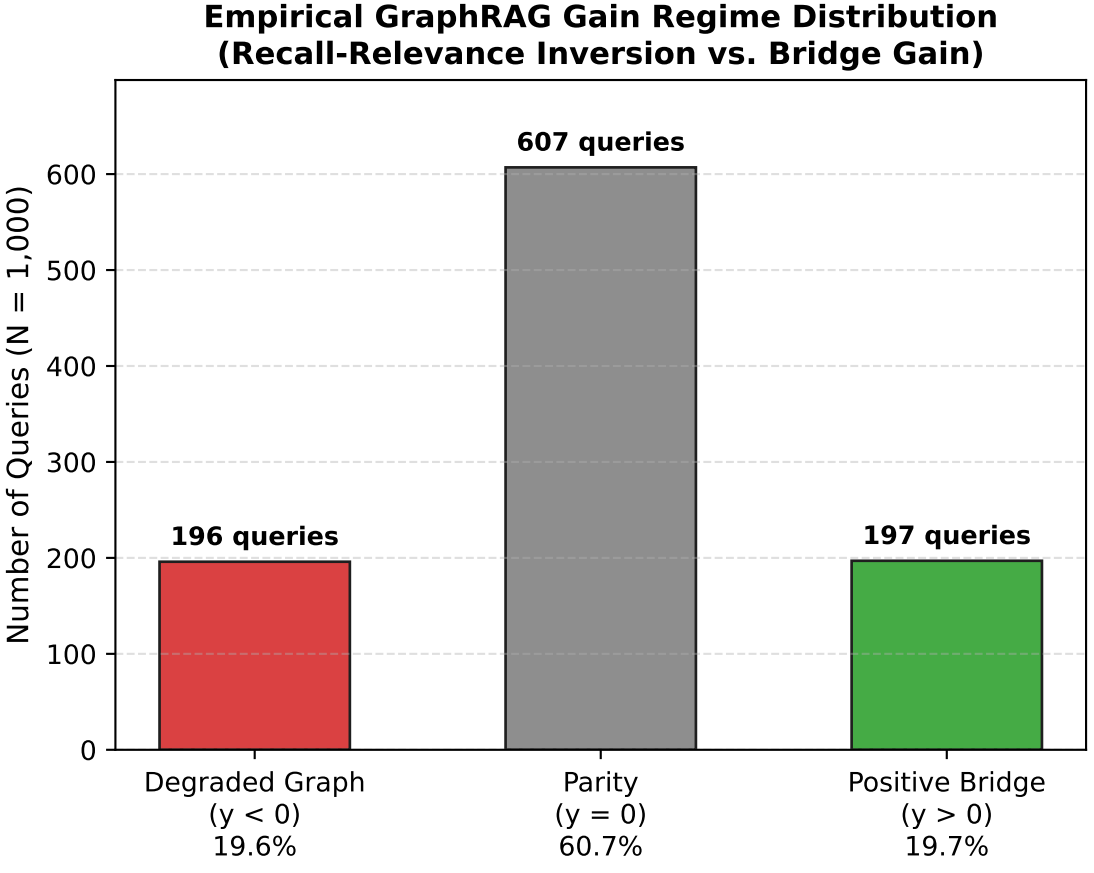
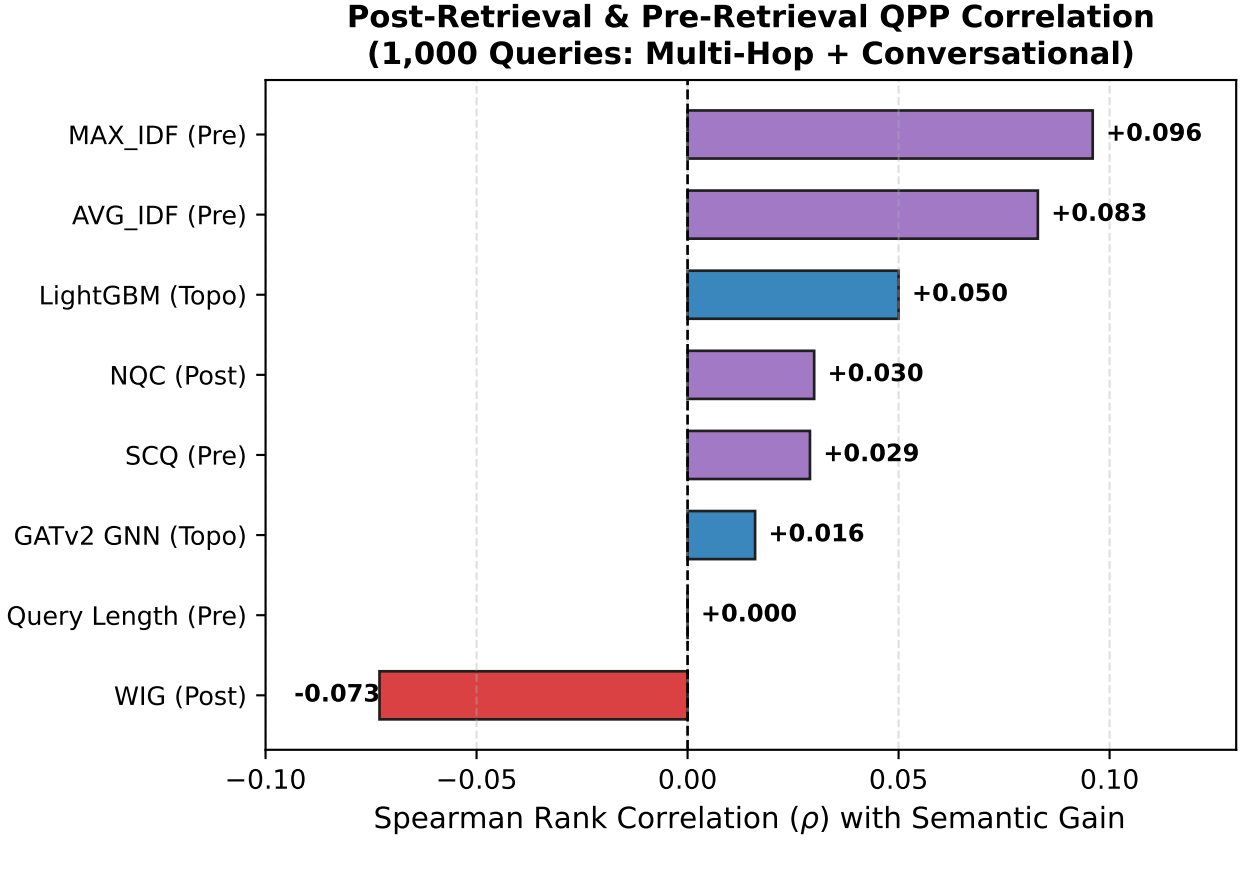

Empirical GraphRAG Gain Regime Distribution
(Recall-Relevance Inversion vs. Bridge Gain)
Number of Queries (N = 1,000)
0
100
200
300
400
500
600
196 queries
607 queries
197 queries
Degraded Graph
(y < 0)
19.6%
Parity
(y = 0)
60.7%
Positive Bridge
(y > 0)
19.7%


Post-Retrieval & Pre-Retrieval QPP Correlation
(1,000 Queries: Multi-Hop + Conversational)
MAX_IDF (Pre)
AVG_IDF (Pre)
LightGBM (Topo)
NQC (Post)
SCQ (Pre)
GATv2 GNN (Topo)
Query Length (Pre)
WIG (Post)
+0.096
+0.083
+0.050
+0.030
+0.029
+0.016
+0.000
-0.073
−0.10
−0.05
0.00
0.05
0.10
Spearman Rank Correlation (ρ) with Semantic Gain